

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Medium

Question

While researching a topic, a student has taken the following notes:

- Generally, an object will heat up when twisted.
- The twisting of an object is known as torsion.
- A 2019 study led by Zunfeng Liu and Ray Baughman tested the torsional heating of various fibers.
- When a 3-millimeter-thick sample of thermoplastic polyurethane (TPU) fiber was twisted, its average surface temperature increased by 6°C.
- When a 4-millimeter-thick sample of styrene-ethylene-butylene-styrene (SEBS) rubber fiber was twisted, its average surface temperature increased by 3.5°C.

The student wants to contrast the two samples. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. When the fibers were twisted as part of the 2019 study, the surface temperature of both samples increased.
- B. In 2019, researchers studied the effect of torsional heating on various fibers, including samples of SEBS rubber and TPU.
- C. Twisting an object will generally cause its temperature to increase, a process known as torsional heating.
- D. The SEBS rubber sample used in the 2019 study was thicker than the TPU sample.

Correct Answer: D

Rationale

Choice D is the best answer. The sentence contrasts the two samples, noting that the SEBS rubber sample, at 4 millimeters thick, was thicker than the TPU sample, which was 3 millimeters thick.

Choice A is incorrect because the sentence mentions a similarity between the TPU fiber and the SEBS rubber; it doesn't contrast the two samples. Choice B is incorrect because the sentence indicates that both the SEBS rubber and TPU samples were part of the 2019 study; it doesn't contrast the two samples. Choice C is incorrect because the sentence describes the process of torsional heating; it doesn't contrast the two samples.